

How Can I Trust AI? : Extending a UXer-AI Collaboration Process in the Early Stages

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ABSTRACT

This paper explores the integration of generative AI into UX design by focusing on developing a collaborative process that facilitates work between UX practitioners (UXers) and AI. It identifies critical barriers in UXer-AI collaboration, investigates the need for modifications in the collaboration process, and proposes a new process involving verification and decision-making stages. Through a workshop with experienced UXers, this study examined their perspectives and challenges in working with AI. The findings emphasize the need to enhance the trustworthiness of generative AI's outputs and discuss various verification methods. This research contributes to bridging the gap between AI technology and practical UX design, paving the way for more effective and trustworthy collaboration.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Generative AI, UX Design, Human-AI Collaboration Process

ACM Reference Format:

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1 INTRODUCTION

Over the past few years, there has been active discussion about the potential and necessity of incorporating AI into the field of UX design [1, 6, 8, 9]. Despite proactive interest and research in academia, most AI tools supporting UX design have not had a significant impact on the industry, and it is said that UX practitioners rarely use them [9, 13]. The reason for this disparity is that the primary focus of research has been merely introducing new AI technologies without progressing to discussions specifically about AI for UX designers [13]. To bridge this gap, this study seeks to

integrate generative AI tools into practical UX design processes. Specifically, our goal is to propose a collaborative process between AI tools and UX design practitioners.

To design a UX designer-specific AI collaboration framework, it is paramount to first thoroughly understand how UX designers interact with generative AI and the challenges they encounter [20]. As generative AI, a more advanced technology than existing AI tools, is still in its early stages of adoption, further discussion is necessary. Research on incorporating generative AI into the design process has tended to prioritize the novelty of the technology over the actual design requirements of users, focusing more on what the new technology can and cannot do [8]. The success of digital tools depends on how well they meet user requirements, so it is crucial to accurately identify these requirements early on and reflect them in the design process [4]. In doing so, these tools can be systematically integrated into UX practice, especially at this juncture, when fundamental changes in the design process are expected due to the introduction of new technological tools.

The research was conducted as follows: Initially, the researcher established a theoretical background for the UXer-AI collaboration process through a literature review. Subsequently, a participatory design workshop was conducted, and task execution among 27 experienced UXers with over six years of experience was observed. Focus group interviews were then conducted in smaller groups of seven to discuss the challenges and requirements when collaborating with AI. Based on the practitioners' feedback, the study identified factors hindering UXer-AI collaboration and outlined implications for strategies to alleviate them.

This paper contributes to the HCI community in the following ways. First, workshops with UX practitioners were conducted to contribute to a deeper understanding of practitioners' perceptions, bridging the gap between research and practice. Second, it provides insights into the three main factors (with eight sub-elements) hindering UXer-AI collaboration and the most significant barriers. Third, it proposes a UXer-AI collaboration process focused on the initial stages, which can implement existing generative AI technologies to mitigate the critical barriers to UXer-AI collaboration.

2 RELATED WORK

2.1 Innovations in UX Design Through Generative AI

Generative AI, which creates outputs (e.g., text or images) based on user-entered prompts, is a highly versatile technology that is expected to be increasingly utilized across various fields [1]. In the realm of UX design, there is active exploration of the potential for AI

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utilization. Examining cases where AI has been integrated into UX design processes reveals that AI can enhance accuracy in research activities, such as problem identification and definition, as well as in solution development tasks, such as interface design, thereby indicating the potential of AI to produce more creative solutions [8, 14, 16]. However, the unique attributes of AI and the changes in the design process resulting from its introduction can slow the innovation of UX design through AI [15]. Since introducing AI into the design process inevitably leads to changes in workflows [14], it is crucial for designers to understand and appropriately utilize the technology. However, due to the complexity of the technology, the unpredictability of the outcomes, the lack of related education, and rapid technological changes, most designers find it challenging to correctly use AI in their work, an issue consistently raised in previous studies [5, 6, 15, 21, 22].

However, as generative AI is still in its nascent stages of adoption, there remains a lack of understanding of the difficulties and requirements designers face when integrating it into various stages of the entire UX design process. Recent research has analyzed and summarized tasks that can be performed using generative AI in the UX design process and those that cannot [8]. However, because the focus has been on introducing the novelty of the new technology and tasks with high potential for automation, generative AI has been applied only partially, not in specialized UX design tasks but rather in routine tasks (e.g., document summarization, presentation slide creation, data organization, or visual tasks for concretizing prototypes) [10, 11]. Furthermore, rather than viewing AI as a perfect replacement for humans, in a collaborative partnership where AI and humans each excel in their respective tasks and complement each other's limitations [19], UXers may face different challenges than before. Therefore, this study proposes a UXer-AI collaboration process that can be integrated across the entire UX design process, comprehensively identifies the factors that hinder this process, and discusses strategies to mitigate UXer-AI collaboration barriers, ultimately enhancing the feasibility of using generative AI in UX design practice.

2.2 UXer-AI Collaboration Process

AI's continuous evolution has led to discussions about the possibility of AI replacing UX designers [8]. Many designers tend to treat AI as a magical tool [5] and blindly trust AI's suggestions or ignore their own judgment, which can infringe on the designer's autonomy [3]. Therefore, a collaborative partnership between humans and AI is essential for the successful integration of AI into the UX design process [14]. In reality, since there are areas in which AI cannot operate, the likelihood of human designers being replaced is low [7]. Hence, UX designers should collaborate with AI by understanding its limitations and leveraging their unique human abilities rather than relying on AI [15, 21, 22].

The UX design process involves understanding user requirements and generating and evaluating ideas [12]. These design processes have a cyclical characteristic, where each stage is repeated to evaluate and review the final solution to ensure the final output meets user requirements. According to Wu [18], the collaboration process between humans and AI is composed of a six-step repetitive process: perceive, think, express, collaborate, build, and test.

Additionally, generative AI, defined as the collaborative agent in this paper, has the flexible characteristic of being able to create new outcomes based on user inputs [13]. In other words, when considering the process in which humans input, AI generates, and then humans input again, it can be anticipated that this collaborative process will also have a cyclical nature before the building and testing of the outcome.

By integrating the UX design process and the AI-human collaboration process, a UXer-AI collaboration process was derived. This process involves a cyclical three-step process of problem recognition, solution derivation, and final result creation, with the characteristic that input and generation tasks are repeated within each stage due to the use of generative AI. By conducting workshops where UXers engage in design work following this process, the aim was to comprehensively understand the factors hindering the UXer-AI collaboration process. Additionally, by discussing effective collaboration strategies, the study sought to improve UXer-AI collaboration experiences and enhance the efficiency of the UX design work process.

As explored in prior research, there is extensive discussion about the potential and methods of using AI in the UX design process. However, due to the lack of AI tools specialized for UX design tasks, AI's implementation in design remains limited [13]. Therefore, it is necessary to comprehensively understand the factors hindering UXer-AI collaboration and practitioners' requirements for successful integration. To support UXers in using AI more effectively in the design process, we asked the following research questions: 1) What are the main barriers for UXers in the UXer-AI collaboration process, and 2) What are the strategies to lower the barriers to UXer-AI collaboration? Ultimately, based on the answers to our research questions, we aim to propose a UXer-AI process that can implement current generative AI technologies in a way that enhances UXer-AI collaboration.

3 METHOD

To observe the process of UXers utilizing AI in their actual work and derive more concrete insights, a UX design workshop featuring generative AI was conducted. The workshop participants consisted of 27 UX designers with at least six years of experience and it lasted approximately 300 minutes. During this process, the participants actively engaged in UX design tasks using generative AI and freely discussed the challenges they faced, points of dissatisfaction, and their requirements.

3.1 Participants

The workshop participants' experience ranged from 6 to 37 years, with an average tenure of 14 years. Regarding their experience with generative AI, all 27 participants had prior experience using it. The most commonly used generative AI tools were ChatGPT, used by 26 participants, and Midjourney, used by 9 participants, with 15 participants having experience with more than two different tools.

3.2 Procedures

The workshop was structured into three parts to identify a range of difficulties and challenges from various perspectives. It included a session to enhance the understanding of generative AI (60 minutes),

Table 1: Workshop (Practical Exercises on UX Design Utilizing Generative AI) Procedures

Participants	27 UX Designers (6 or more years of experience)		
Steps	Education (60 minutes)	Task performance (180 minutes)	Focus group interview (60 minutes)
Contents	Introducing concepts and usage methods to enhance understanding of generative AI	1. Recognize problem (text-to-text AI) (text-to-text AI) 2. Derive solutions (text-to-text AI) 3. Generate the final outcome (text-to-image AI)	1. Difficulties 2. Applicability 3. Concerns

Table 2: Eight Factors that Hinder UXer-AI Collaboration

Category	Element	Content
Challenges arising from unfamiliarity with AI	Prompt input	<i>"In the end, how you use the prompt is the most crucial. Using the prompt is challenging."</i>
	Generate	<i>"Even when using GPT-4, the specificity remains limited"</i> <i>"I'm unsure how my requirements have been reflected on the screen."</i>
	Tool selection	The participants, before undertaking the task, asked, <i>"Which tools are best to use?"</i> After completion, they discussed the optimal combination of programs.
Issues requiring internal support within the company	Financial	<i>"Since all the truly usable tools are paid, utilizing a paid AI for the task would have been more beneficial."</i>
	Security	When using the internal corporate network, there is an issue with access to a specific AI service being blocked.
Critical barriers in UX design tasks relying on research data	Hallucination	<i>"Since there is no back data provided for the answers, a verification process is needed to ensure they are not hallucinations."</i>
	Reliability	<i>"GPT doesn't provide sources, so how can we verify this?"</i>
	Decision	<i>"In the absence of source data, it becomes necessary to perform tasks like A/B testing to filter out trustworthy responses."</i>

a task performance session using various generative AI tools in UX design (180 minutes), and focus group interviews (60 minutes).

In the first part, considering that some participants were unfamiliar with generative AI, the concepts and methods related to the use of generative AI were explained. The task performance in the second part was conducted according to the UXer-AI collaboration process. The task was unified as a redesign of the Netflix mobile app, considering it to be a representative global over-the-top (OTT) service with sufficient data for generative AI to have learned. The detailed tasks included using text-generating AI tools (e.g., ChatGPT, wrtn, Google Bard, and Naver Clova X) to analyze current service problems, derive solution ideas for the defined problems, and create prototype designs for the solutions using Blue Willow, Stable Diffusion, and the Figma AI plugin.

Participants recorded their thoughts on the communication support tool Padlet after completing each task. After completing all tasks, the workshop participants were divided into four groups of seven each for focus group interviews. During this process, participants discussed the content recorded on Padlet, the difficulties experienced during task performance, potential applications in practice, and concerns. The researcher took turns visiting each group for 15 minutes to transcribe the discussion content.

3.3 Data Analysis

We conducted qualitative analysis using thematic analysis [2] based on the transcriptions to identify themes related to challenges and requirements in the UXer-AI collaboration experience. The transcriptions included verbal comments made by participants during

the workshop, opinions recorded on Padlet, and content transcribed by the researcher during the focus group interviews. To identify themes and patterns, we utilized Atlas.ti, a qualitative analysis tool.

Initially, we read the raw data and performed open coding. Following this, we reviewed and developed the initial codes obtained from this process and classified them into higher-level themes. Finally, through discussion, we further refined and concretized subcategories for each of the higher-level themes.

4 FINDINGS

4.1 Challenges Due to Unfamiliarity with AI

In the case of prompts, the type of output varied depending on how prompts were entered, causing participants to spend a lot of time figuring out the correct way to input prompts, often asking, "How should I input this?" during task execution (P2, 13, 19). This difficulty extended to the generation of outputs. On Padlet, comments such as "Even using GPT-4, the latest data isn't reflected, and it often just lists general problems of streaming apps, lacking specificity" and "It's too hard to generate outputs as I intend" were recorded (P20, 22). These challenges make selecting the appropriate AI tool difficult, as the quality of the outputs varies depending on which AI is used. However, as participants became familiar with the characteristics of various AIs while performing tasks, they began to record comparisons of AI tools' traits on Padlet (P9, 15, 16, 18, 21), such as "ChatGPT vs. Google Bard: For the same question, ChatGPT's answer is much more structured and analytical." Therefore, as UXers become more accustomed to using AI, it is expected that

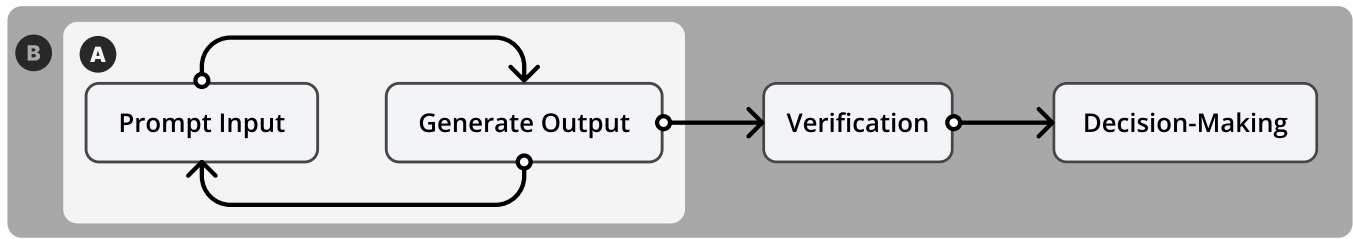


Figure 1: A(Two stages) : The initial UXer-AI collaboration process, B(Four stages) : The newly proposed process

these difficulties will diminish, and the degree to which all three challenges are resolved will increase.

4.2 Issues Requiring Internal Company Support

Some participants had subscribed to paid services for personal use, while others only used free AI tools for their tasks. A notable difference in the quality of the outputs between these two groups was observed. Following the workshop, one participant commented, “It seems that the tools that are genuinely useful are all paid, so working with paid AI would have been more helpful,” expressing disappointment with the free AI tools (P26). Additionally, as the workshop was conducted using the company’s internal network, there were security issues that led to the blocking of access to LimeWire, an image-generating AI platform. This raised concerns among the participants about using such tools in a work environment. These two issues suggest that internal support from companies, such as providing AI services that are behind a paywall and establishing AI usage policies that consider company security, could be beneficial.

4.3 Barriers Crucial to UX Design Work

Generative AI has the issue of hallucination, where it indiscriminately collects information and generates persuasive answers regardless of their truthfulness [11]. A participant (P16) pointed out this issue: “Since back data is not provided for the answer contents, a verification process is necessary to ensure they are not hallucinations.” Some participants questioned the reliability of the data (P3, 12, 14), asking, “GPT doesn’t provide evidence; how can I verify this?” Furthermore, on Padlet, concerns about data reliability were expressed with remarks such as, “Google Bard tends to reflect more recent data, but the trustworthiness of its output still seems inadequate.”

The issues of hallucination and reliability in responses led to difficulties in decision-making, as it is challenging to filter out trustworthy answers without supporting evidence. In light of this, one participant wrote on Padlet that direct tasks, such as A/B testing, should be carried out to facilitate informed decision-making (P19). In UX design work, the final outcome varies depending on how the designer sets the direction, making decision-making crucial. However, the use of generative AI, which generates various responses without confirmed validity, further accentuates the problem of decision-making for UXers.

5 DESIGN IMPLICATION

The research findings highlight that mistrust in the outputs generated by AI posed the most significant obstacle to UXer-AI collaboration. However, verifying AI’s operations and providing sources for its outputs are technically challenging [17]. Consequently, a secondary validation process is necessary, and this discussion during workshops revolved around methods that UXers found effective. Therefore, this study aimed to reformulate the UXer-AI collaboration process based on these methods (Figure 1, 2).

5.1 Four-Stage Process: Input, Generation, Verification, and Decision-Making Mode

Traditional generative AI focuses solely on the tasks of input and generation. This means that without a separate verification procedure for responses, UXers find it difficult to assess the reliability of AI-generated answers, complicating the decision-making process. Even when the need for verification is recognized, users utilizing conventional generative AI may easily overlook the verification process and remain unaware of how to conduct it. Practitioners who attended the workshops spoke of the need to verify AI responses, yet they often struggled with actual methods of verification.

The four-stage process described in this section, consisting of input, generation, verification, and decision-making tasks, is structured differently from the previous two-stage process, allowing for the sequential execution of these tasks (Figure1). In the generative mode, the same tasks as those executed by conventional generative AI are performed, and in the verification mode, three types of verification tasks can be executed. This is based on feedback from practitioners in workshops who mentioned that these methods were helpful in enhancing trustworthiness and resolving hallucinations. In the final decision-making mode, users can access answers that have been prioritized, summarized, and organized based on the results of the verification process.

5.2 Source: A Method to Verify If There Is a Match with the Data Directly Trained by the User

During the focus group interviews, all subgroups emphasized the need for AI to provide evidence. They highly rated AI tools, such as wrtn or Bing AI, offer sources for their answers along with website links. Especially in practice, when drafting reports, the necessity to cite sources means that wrtn’s feature of providing sources could serve as a solution, a view that garnered widespread agreement (P21). However, in the corporate UX design context, research has

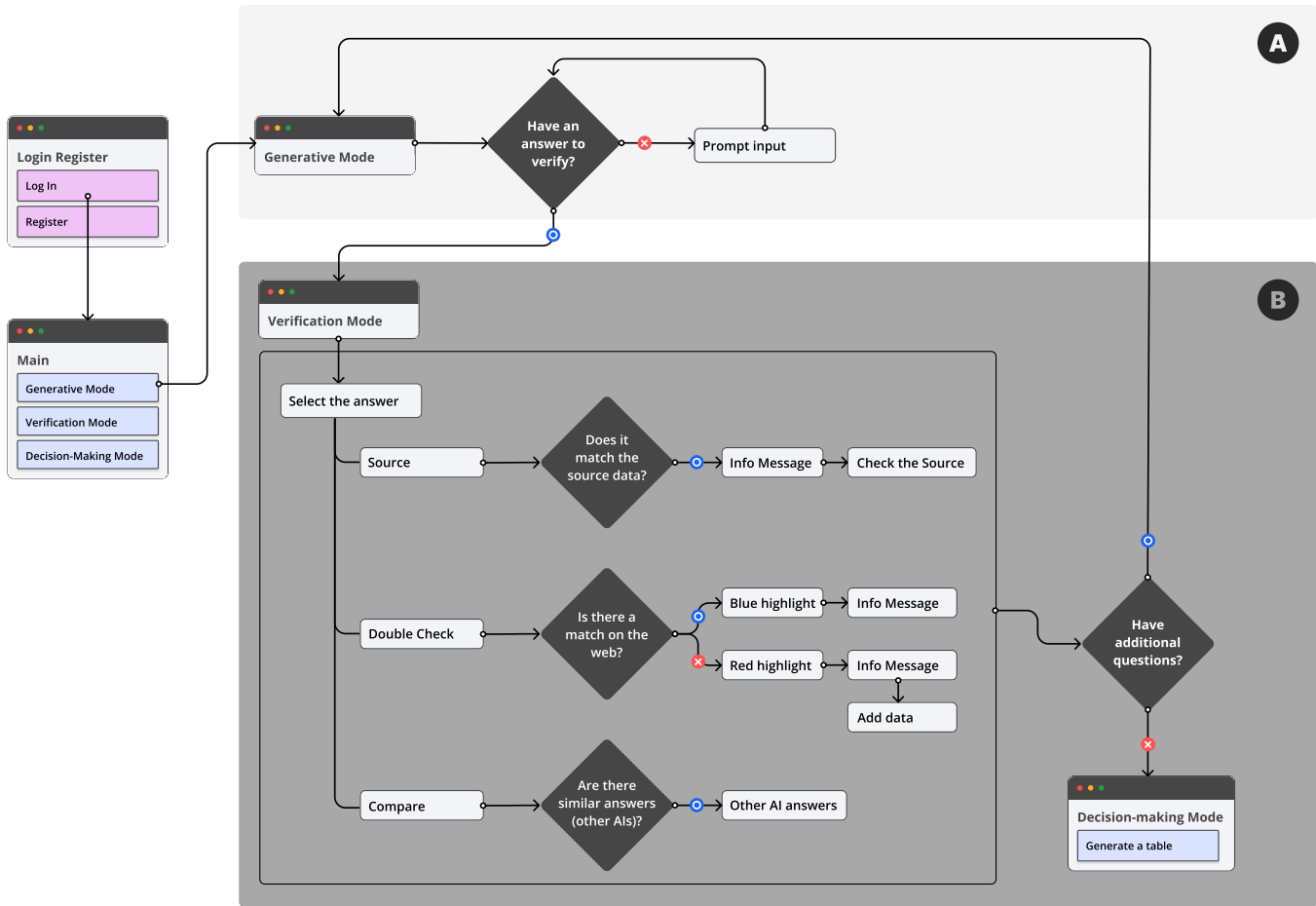


Figure 2: Flow chart: The existing UXer-AI collaboration process merely repeats the cycle of UXers inputting prompts and AI generating responses (A). The new process allows for decision-making based on the results of verifying the generated responses through three different methods (B). Additionally, UXers with further questions can return to the initial stage, the generation mode.

often targeted specific services and audiences, meaning that the required data are more likely to be held internally rather than on the internet. Considering the technological advancements that now allow users to train AI tools with their own data and subsequently extract information from such data, implementing a feature that compares AI-generated responses with user-trained data to identify similarities could significantly enhance the utility of AI for UX practitioners. Additionally, internal materials, such as usability tests and user reviews, can serve as evidence for the responses, further increasing the reliability of the data.

5.3 Double Check: A Method to Search for Similar Content on the Web

Double check is a feature derived from Google Bard’s “Google it” function. This function re-Google the AI-generated answers to determine if similar content is available on the web. Participants utilized this feature during task execution and recorded positive responses on Padlet (P6, 19), such as “Using Bard is great as it

provides evidence” and “Google Bard’s hallucination verification feature was helpful in decision-making.” However, some noted that there were still many answers that could not be verified (P4, 12). This indicates the need to employ a variety of verification methods in parallel rather than relying on just one.

5.4 Compare: A Method of Comparing Answers from Various AI Tools (Assistance in Decision-Making)

Even with the same prompt, responses can vary depending on which generative AI tool is used. While some participants found it challenging to make decisions due to the differing responses from the AI tools they used, others who discovered commonalities between the responses of various AIs stated that it was helpful for decision-making (P18, 25). For example, one participant (P18) noted, “When I asked a question, both GPT and Bard highlighted ‘Lack of personalization’. This really helped in making a decision.” However,

currently, there is the inconvenience of having to access various AI tools individually and repetitively input and generate the same prompts. If improvements could be made to enable simultaneous review and comparison, similar to the compare function, participants would likely find it easier to make decisions.

5.5 Limitations and Future Work

This paper discussed verification methods centered around tasks that workshop participants mentioned as effective for enhancing trustworthiness, proposing an integrated UXer–AI collaboration process. Considering the rapid advancement of generative AI technology, the strategy of assessing only currently available technologies could be a limitation of this research. Moreover, since the discussion was limited to participants, it is necessary to further conduct workshops to examine the effectiveness of these methods among a broader range of UX professionals.

The verification methods discussed in this section are applicable to the problem analysis and solution derivation stages, and different verification methods, such as usability tests and user interviews, would be needed when deriving final outcomes. Future research could extend the discussion to include verification methods when using image and prototype generation AI, thereby applying verification throughout the entire UX/UI design process.

6 CONCLUSION

This study investigated the factors hindering collaboration between UXers and AI, with a focus on those most detrimental to UX design tasks. Workshops and focus group interviews were conducted with 27 UX practitioners who engaged in collaborative tasks with generative AI, and their opinions were qualitatively analyzed. The findings revealed that UXers generally faced eight barriers when collaborating with AI. These can be categorized into three groups: issues due to unfamiliarity with AI, problems requiring internal company support, and critical issues in UX design tasks.

Among these, it was discovered that the root cause of the most critical barriers lies within the UXer–AI collaboration process. The existing process, which produces unverified, repetitive answers, leading to concerns about data reliability and hallucinations, and complicating decision-making. Therefore, this study proposed a UXer–AI collaboration process to enhance reliability by adding verification and decision-making stages to the existing process. The outcomes of this study offer design implications for UX design-specific AI systems aimed at improving the UXer–AI collaborative experience. This research contributed to a more comprehensive understanding of the barriers that need to be overcome for effective UXer–AI collaboration in practice and provided insights into a future direction for the development of the UXer–AI collaboration process.

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A EXTENDED OVERALL PROCESS DIAGRAM

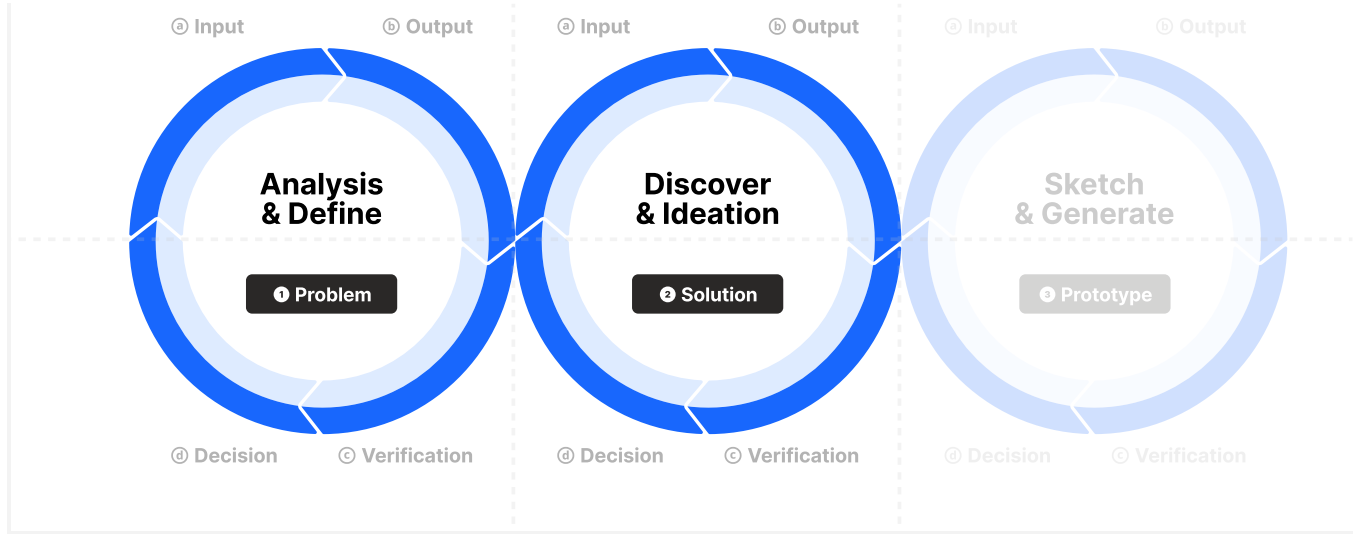


Figure 3: Repetition of a)Input, b)Output Generation, c)Verification, and d)Decision-Making Tasks in the Early Stages (1-Problem Recognition, 2-Solution Derivation) of the UXer-AI Collaboration Process)

B PROTOTYPE FOLLOWING THE NEWLY UXER-AI COLLABORATION PROCESS

Mode

Generative mode

Verification mode

Decision-making mode

Version: GPT 3.5

Show resource link

Show proposed prompt

Dark mode

Trustworthy AI

Help Guide

User 10232

It'd be nice to redesign UI of Netflix. Can you select one problem that I need to redesign?

Source

Double-check

Compare

Need to check

How doing

1. Customization and Personalization

- Explore ways to allow users to personalize their profiles further, such as choosing different themes or background images.
- Consider introducing customizable avatars or profile pictures to make each user's profile unique.

2. Content Recommendations

- Enhance the profile settings to provide more granular control over content recommendations. Users may want to fine-tune their preferences directly within the profile settings.

3. Visibility and Access

- Evaluate how easily users can switch between profiles and make adjustments. Streamlining this process can improve the overall user experience.
- Consider implementing quick profile switching options or a dashboard that provides an overview of each profile's viewing history and preferences.

4. Parental Controls

- If not already in place, consider improving parental controls within profiles. This could include more robust content filtering options and the ability to set time limits for certain profiles.

<Priority Ranking Based on Weighting>

1. Presence of directly trained data (Source)
2. Existence of similar content during web exploration (Double-Check)
3. Presence of similar content when compared with responses from other text-generating AIs (Overlap)

Contents	Source	Double-Check	Overlap
1. Parental Controls: Consider improving parental controls within... more	Survey_20230219.doc →	www.acm.library.com	CLOVA X Google Bard Bing AI
2. User Feedback: Implement a feedback mechanism within the profile settings... more	Usability Test_20230510.doc →	www.acm.library.com	Google Bard Bing AI
3. Contents Recommendations: Users may want to fine-tune their preferences... more	User review_20220729.doc →		

1 Verification Mode & Mode menu(left)

This is the screen for selecting responses generated in the generative mode for verification. Once a response is chosen, it can be verified using 3 methods.

2 Decision-making mode

In the Decision-making mode, responses are organized based on three verification tasks. UXers can rely on this table for decision-making support.

in the profile settings.

Source

Double-check

Compare

Need to check

How doing

as choosing different themes or background images.

Consider introducing customizable avatars or profile pictures to make each user's profile unique.

Content Recommendations

Enhance the profile settings to provide more granular control over content recommendations. Users may want to fine-tune their preferences directly within the profile settings.

Visibility and Access

Evaluate how easily users can switch between profiles and make adjustments. Streamlining this process can improve the overall user experience.

Consider implementing quick profile switching options or a dashboard that provides an overview of each profile's viewing history and preferences.

4. Parental Controls

- If not already in place, consider improving parental controls within profiles. This could include more robust content filtering options and the ability to set time limits for certain profiles.

1 Verification mode : Source

If there are portions in the generated response that match pre-learned data, it is highlighted in blue, and a message is displayed indicating the data name.

2 Verification mode : Double Check (Blue)

If there are parts in the generated response that match web search results, it is highlighted in blue, and a message is displayed indicating the link.

3 Verification mode : Compare

When different generative AIs provide similar responses to the same prompt, you can compare the answers to see what each AI has generated.

Figure 4: Prototype : 1) Full screen (with three mode menus on the left) 2) Decision-making mode table: Organizing answers in order of reliability based on verification work 3) Guidance message for answers verified by source 4) Guidance message for answers verified by double check 5) Various AI responses viewable through the compare feature